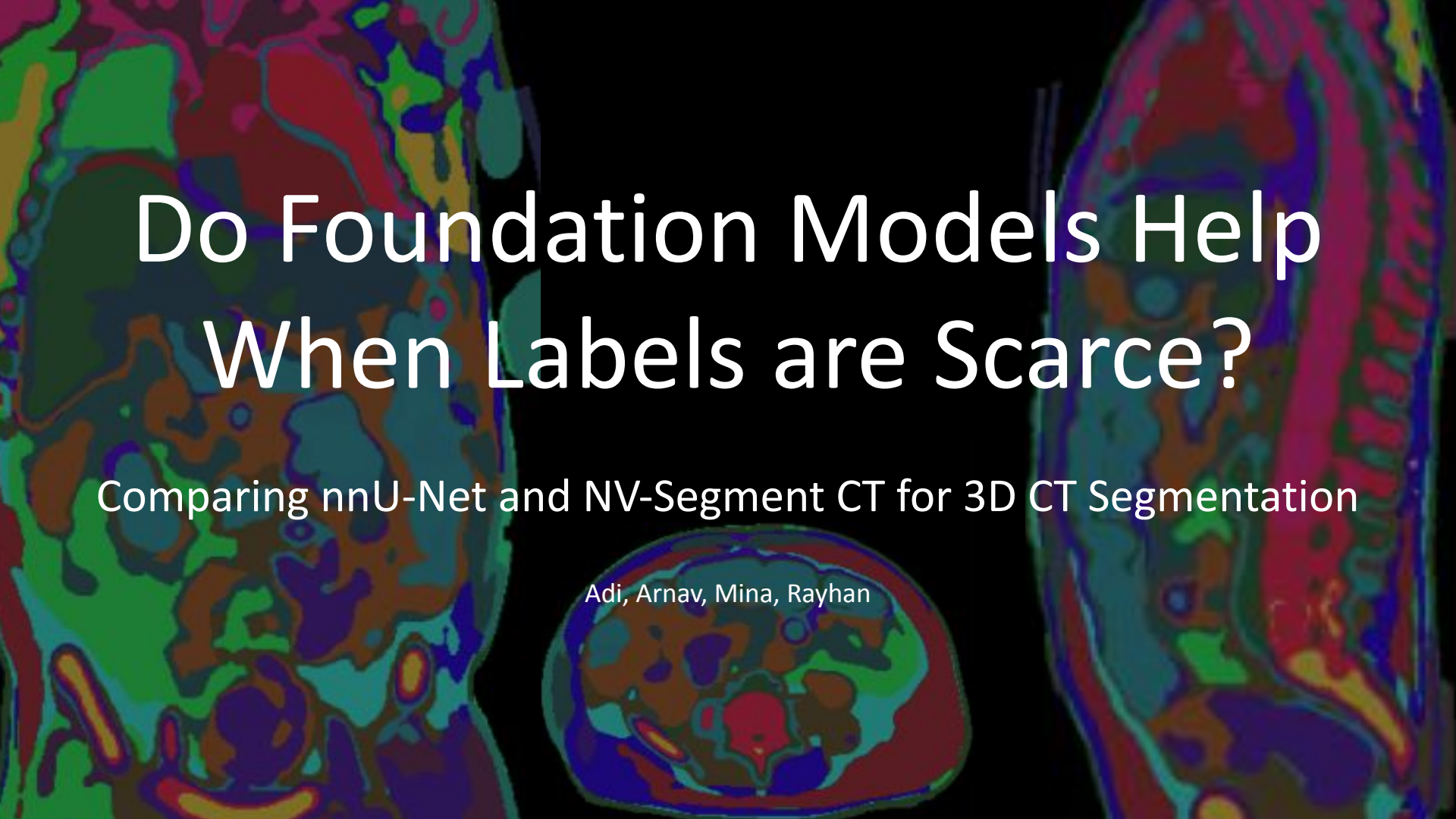
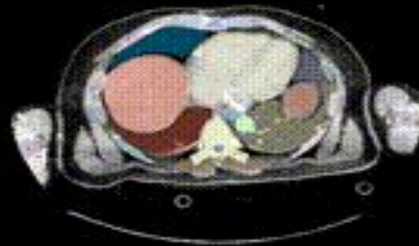
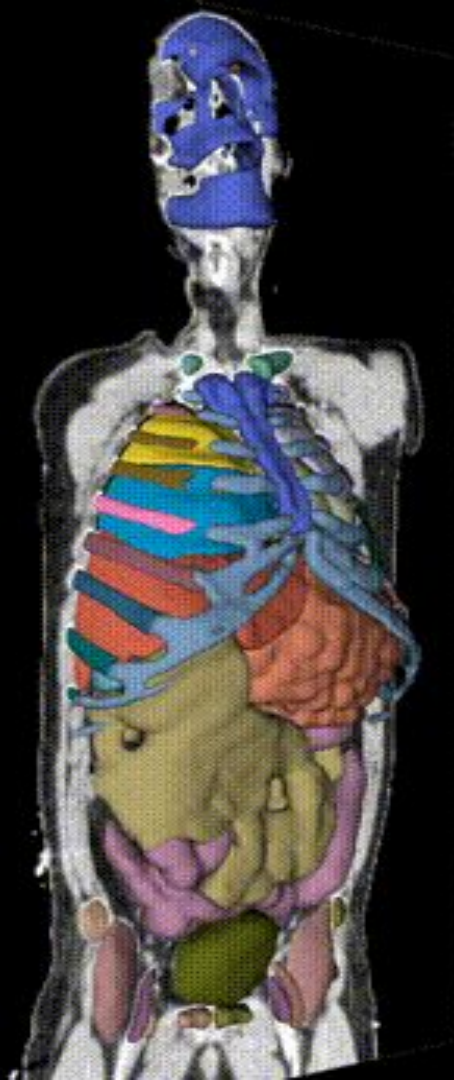


Do Foundation Models Help When Labels are Scarce?

Comparing nnU-Net and NV-Segment CT for 3D CT Segmentation

Adi, Arnav, Mina, Rayhan



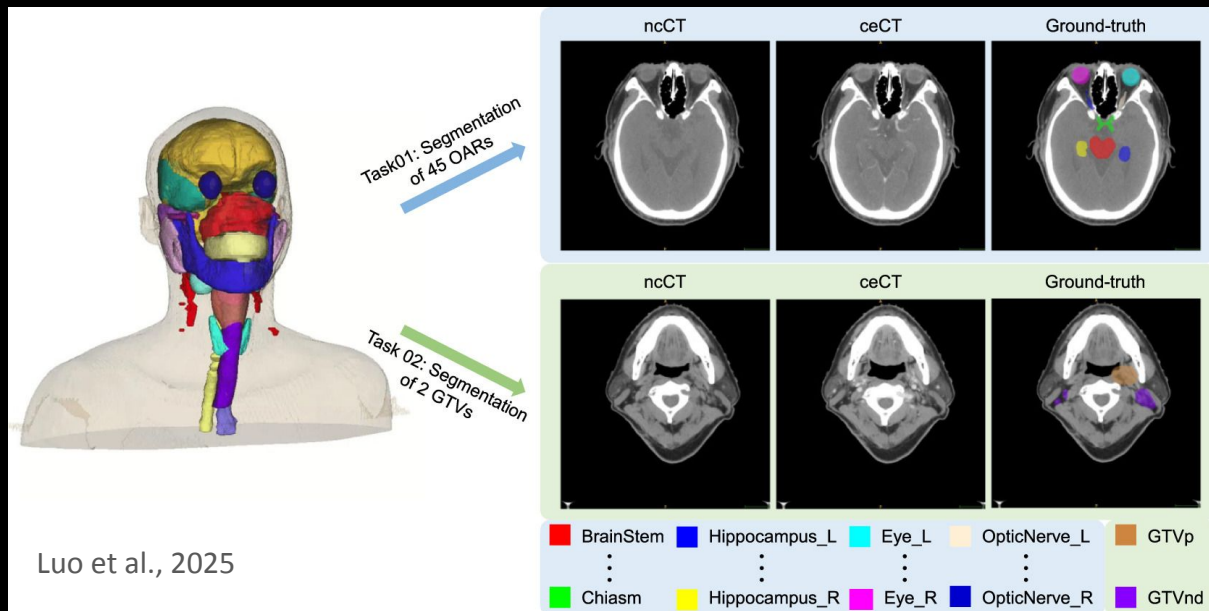


3D segmentation requires **voxel-level** labels, but expert annotation is **costly** and **time-consuming**.

Can **pre-training** reduce annotation needs?

- 1 Can **pre-training** reduce annotation needs?
- 2 When does pre-training **outperform** training from scratch?
- 3 Does the advantage **decrease** as labels **increase**?
- 4 Does the **distribution** of available labels matter?

SegRap2023 Dataset: **200** patients with CT scans, **120** labeled patients for training, **45 organ** expert labels (+ 2 tumor volumes)

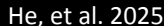


Target output: 3D anatomical segmentation masks + labels

- Task-specific segmentation model
- Trained from random initialization
- Automatically configures the training pipeline



- Pretrained 3D CT foundation model
- Begins with learned anatomical representations
- Fine-tuned using SegRap2023 labels

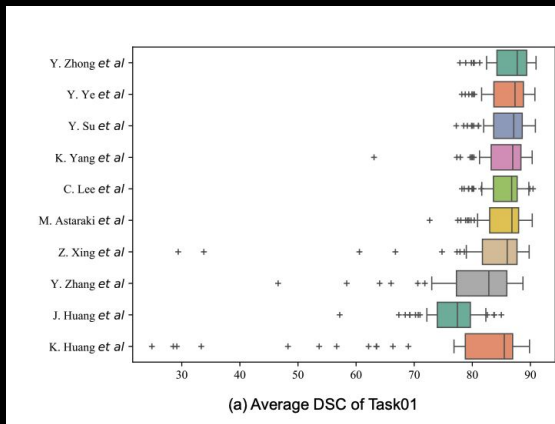


VISTA3D showed **stronger** performance on abdominal CT (WORD) with **fewer** cases, but this performance gap **narrowed** as labeled cases increased.

Table 3. Finetuning performances of average test dice scores, with respect to the number of training cases.

Dataset	# of cases	Auto3DSeg	nnU-Net	TotalSegmentator	VISTA3D
Micro-CT Mouse	1	0.820	0.759	0.791	0.926
	5	0.923	0.922	0.924	0.935
	10	0.934	0.930	0.936	0.938
	20	0.947	0.942	0.944	0.944
	40	0.949	0.949	0.949	0.948
	89	0.949	0.949	0.951	0.951
WORD	1	0.214	0.185	0.779	0.795
	5	0.611	0.562	0.823	0.839
	10	0.744	0.697	0.837	0.855
	20	0.806	0.793	0.855	0.862
	40	0.862	0.831	0.857	0.869
	100	0.873	0.874	0.875	0.875

He et al., 2025



SegRap2023 solutions were **largely nnU-Net-based**, but **pretrained** universal models such as UniSeg were also competitive.

Luo et al., 2023

1 **Data validity:** Verify SegRap2023 was not used during pre-training of NV-Segment-CT

2 **Model adaptation with fully labeled patients**

- Select 5, 10, 20, 40, or 80 patients
- Keep all target structure labels
- Train nnU-Net from scratch + Fine-tune NV-Segment-CT
- Compare on the same fixed test set

3 **Model adaptation with partially labeled patients**

- Control the overall annotation budget
- Select number of patients and number of structure labels (e.g. 10 patients x 45 structures vs 90 patients x 5 structures)
- During training, missing structures treated as unknown, not background
- Compare with the fully labeled setting

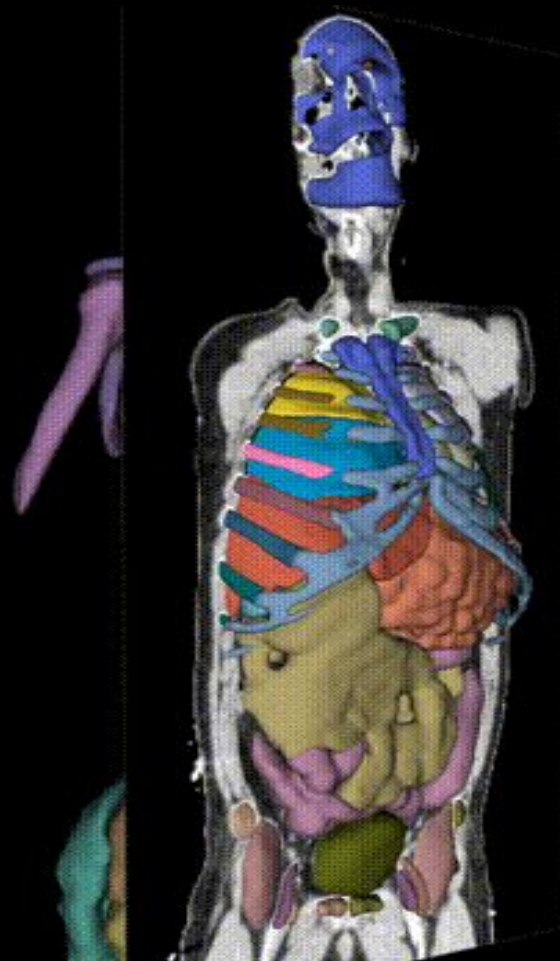
Compute Requirement

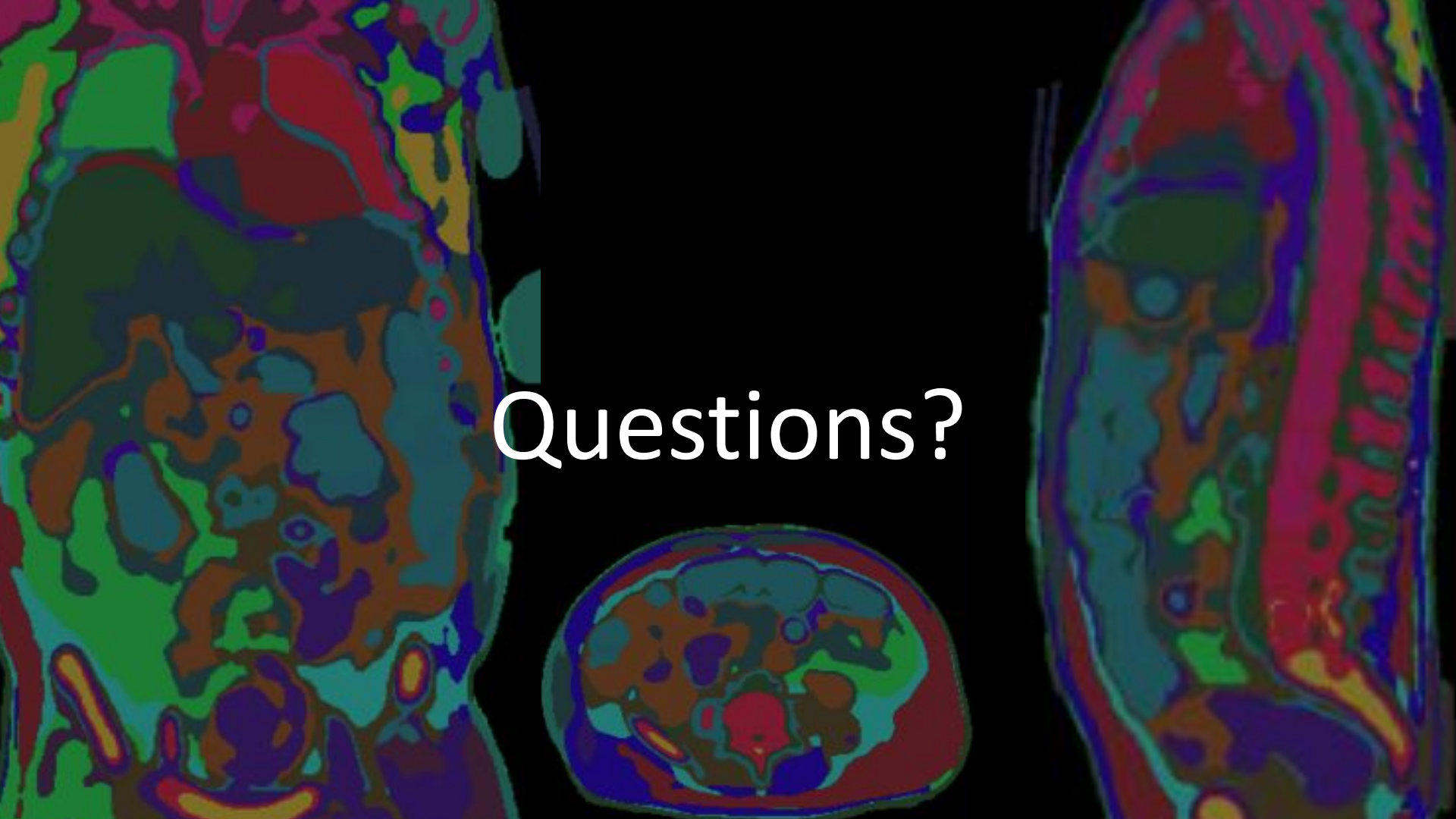


- **GPU cluster likely required**
- nnU-Net: ~16–32 GB GPU memory
- Foundation-model fine-tuning: ~32–48+ GB useful

Expected Insights

- Determine when **pre-training** provides the largest benefit
- Determine whether labels are better spent **densely or sparsely**
- Identify the most **effective strategy** under limited annotation resources



The image features three MRI scans of a human torso, likely a female, showing internal organs. The scans are color-coded: red for soft tissue, green for fat, blue for fluid, and yellow for bone. The top-left scan is a coronal view, the top-right is a sagittal view, and the bottom-center is an axial view. The word "Questions?" is written in white text across the center of the image.

Questions?

First Questions We Will Investigate

- **Data validity:** confirm no SegRap2023 overlap with foundation-model pretraining
- **Label budget:** compare fully labeled patients vs. partially labeled patients
- **Model adaptation:** compare nnU-Net training from scratch with NV-Segment-CT fine-tuning
- **Fine-tuning strategy:** test partial vs. full fine-tuning across label budgets

Pre-training and Data Checks

- Verify SegRap2023 was not used during pre-training
- Check documented VISTA3D training datasets
- No documented SegRap2023 overlap found
- Identify SegRap structures represented in NV-Segment-CT
- Map compatible labels between the two datasets

Experiment 1: Fully Labeled Patients

- Select 5, 10, 20, 40, or 80 patients
- Keep all target structure labels
- Train nnU-Net from scratch
- Fine-tune NV-Segment-CT
- Compare on the same fixed test set

Experiment 2: Partial Labels

- Use CT scans from a larger patient pool
- Keep only selected structure labels per patient
- Missing structures treated as unknown, not background
- Control the overall annotation budget
- Compare with the fully labeled setting

Overall Experimental Setup

- Fixed training / validation / test split
- Compare nnU-Net vs. NV-Segment-CT
- Two label-scarcity settings
 - Fully labeled patients
 - Partially labeled patients
- Label budgets: 5, 10, 20, 40, 80
- Repeat sampling to measure variability
- Compare two interpretations of label scarcity

First Questions to Investigate

- Effect of label budget
- Effect of label distribution
- Partial vs. full fine-tuning
- Foundation model vs. train-from-scratch baseline

Fine-Tuning NV-Segment-CT

Partial Fine-Tuning

- Freeze most pretrained layers
- Train task-specific layers
- Lower risk of overfitting with few labels

Full Fine-Tuning

- Update the entire model
- Use a lower learning rate
- Test across the same label budgets

Evaluation

Accuracy

- Dice score — segmentation overlap
- HD95 — boundary error

Stability

- Variation across repeated patient samples

Failure Modes

- Missed structures
- False positives / boundary leakage

Cost

- GPU memory
- Training and inference time